

Program : Diploma in Textile Technology	
Course Code : 5063C	Course Title: Long staple spinning Technology
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To enable the students to understand the method of yarn production from long staple fibres like wool and flax using long staple spinning technology.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basics of Long Staple Fibres		Textile Fibers	2
Knowledge of short staple spinning preparation process		Yarn Manufacture I	3
Knowledge of short staple spinning		Yarn Manufacture II	4

Course Outcomes

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Discuss the principle of opening, cleaning and carding of long staple fibers.	15	Understanding
CO2	Describe the principle and working of drawing machine suitable for long staple fibers	10	Understanding
CO3	Discuss the wool combing operation , worsted top finishing and roving formation process employed in Long staple spinning industry	18	Applying
CO4	Illustrate the different methods of Long staple Yarn spinning and summarize the yarn manufacturing process. Flow chart for long staple fibres.	15	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	2					
CO3	3	2					
CO4	3	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Levels
CO1	Discuss the principle of opening, cleaning and carding of long staple fibers.		
M1.01	Explain the basics of Long staple fibres and the method of removing impurities in the wool fibre.	3	Remembering
M1.02	Discuss the methods adopted to process raw flax & jute and Automatic fibre blending for the woollen system	4	Understanding
M1.03	Describe the working principle of carding machine and details of different type of carding machine-worsted carding, semi – worsted carding, woollen carding	4	Understanding
M1.04	Illustrate the working of Flax carding and jute carding.	4	Understanding
Contents: OPENING, CLEANING AND CARDING Introduction to long staple fibres - Raw wool processing - Impurities in the wool fibre and their removal - Wool opening and shaking - Wool scouring - Raw Flax Processing - Raw jute processing - Automatic fibre blending for the woollen system - Objectives of carding-working principle and details of different type of carding machine - worsted carding, semi - worsted carding, woollen carding, flax carding and jute carding			
CO2	Describe the principle and working of drawing machine suitable for long staple fibers		
M2.01	List the Objectives of drawing and Principle of long-staple drafting	2	Remembering
M2.02	Explain the methods of fibre control in long staple drafting zone	3	Understanding

M2.03	Discuss the working of worsted, semi worsted drawing	2	Understanding
M2.04	Describe the working of jute and flax drawing	3	Understanding
	Series Test - I	1	
Contents: DRAWING PROCESS Objectives of Gilling process - preparative - Principle of long - staple drafting - fibre control in long staple drafting zone - Pin control, Ambler draft - sliver doubling - Description of worsted, semi worsted, jute and flax drawing - Applications of Autolevelling			
CO3	Discuss the wool combing operation , worsted top finishing and roving formation process employed in Long staple spinning industry		
M3.01	Discuss the basic principles of wool combing and the comber preparatory processes involved.	4	Remembering
M3.02	Describe the general features of a rectilinear wool combing machine and the wool combing cycle sequence of operations	4	Understanding
M3.03	Explain the principle of Twist insertion in roving frame and the concept of Flyer lead delivery	5	Understanding
M3.04	Illustrate the operating principle of roving machine.	5	Understanding
Contents: COMBING AND ROVING FRAME Objective of combing - basic principles of wool combing - The rectilinear wool comb Preparatory processes - General features of a rectilinear wool comb - wool combing cycle of operations - worsted top finishing - Finisher gilling and packaging - Re - combing Expressing yield and noil production in combing - Tear ratio, Romaine(%) and Noil (%). Twist insertion in roving frame - Flyer lead delivery - Operating principle of roving machine.			
CO4	Illustrate the different methods of Long staple Yarn spinning and summarize the yarn manufacturing process flow chart for long staple fibres.		
M4.01	Discuss the different input material fed to wool ring frame and the working mechanism of woollen Ring spinning frame	3	Understanding

M4.02	Explain the working mechanism of Worsted spinning frame and Semi worsted spinning frame	4	Understanding
M4.03	Describe the Rings and Traveler used for woolen and flax spinning.	4	Understanding
M4.04	Illustrate the working of Precision winder and illustrate the Process flow chart for manufacturing woolen, worsted, flax and jute	4	Understanding
	Series Test - II	1	

Contents:

YARN SPINNING AND WINDING

Difference between woolen yarn and worsted yarn - Different input material fed to wool ring frame - working mechanism of woollen Ring spinning frame - Drafting and the false twist device - Worsted spinning frame - Semi worsted spinning frame - Double zone drafting system - Lubricated conical rings - J shaped travelers - Flax spinning Ring and Traveller - self - twist spinning system - Yarn winding - Precision winders

Process flow chart for manufacturing woollen, worsted, flax and jute

Text / Reference:

T/R	Book Title/Author
T1	Oxtoby E., "Spun Yarn Technology", Butterworths, London,
R2	Richards RTD and Sykes A B," Woollen Yarn Manufacture", The Textile Institute, Manchester 1994.
R3	Sharp P, "Flax, Tow and Jute Spinning", Abhishek Publications, Chandigarh, 1998
R4	Stout H P, "Fibre and Yarn Quality in Jute Spinning", The Textile Institute, Manchester, 1988.
R5	Chernysheva N, "Wool Spinning" Vol 1&2, Mir Publishers, Moscow, 1983.

Online resources:

Sl.No	Website Link
1	https://woolery.com/modern-flax-hemp-and-jute-spinning-and-twisting.html
2	https://www.woolmark.com
3	https://www.bracker.ch
4	https://www.woolwise.com/wp-content/uploads